Gqibindlala Catering & Tent Hiring

Online Booking & Inventory System — Project Brief

The Idea

Gqibindlala Catering & Tent Hiring rents out event equipment — tents, chairs, tables, a mobile fridge, and a meat cutter — for functions and gatherings. Today, customers likely enquire by phone or in person, and availability is tracked manually. This project builds a web system that lets customers browse equipment, see real availability, and submit a booking themselves, while giving the business a live view of stock, bookings, and payments in one place.

What It Is

A web application with two sides: a public-facing site for customers, and an admin dashboard for the business.

Public Site (Customers)

- Browse equipment by category — chairs, tables, tents, mobile fridge, meat cutter
- Pick a quantity and a start/end date range for each item added to a booking
- See real availability — dates already hired and paid for are hidden per item
- Submit an enquiry/quote, which becomes a pending booking

Admin Dashboard (Business)

- View and manage all incoming bookings (pending, confirmed, cancelled)
- Manage the equipment catalogue — quantities, pricing, photos
- Confirm a booking once payment is received, which then blocks those dates
- Get an overview of upcoming orders and equipment on hire

How It Works

Each item a customer adds to a booking is tracked on its own — separate quantity and its own start/end date range — rather than one shared date range for the whole order. This matters because a customer might need chairs for two days but a tent for five, within the same booking.

Booking flow:

- Customer selects equipment, quantity, and a date range for each item
- System checks availability — quantity-aware, so items are only fully blocked once the remaining stock for those dates reaches zero
- Customer submits the booking, which starts as **pending**
- Business confirms the booking once payment is received, marking it **confirmed**
- Once confirmed, that item's dates are removed from availability for other customers

Pending (unpaid) bookings do not block the calendar — availability is only locked in at payment confirmation, on a first-confirmed basis.

Tech Stack

Layer	Technology	Purpose
Backend	Java — Spring Boot	REST API, business logic, security
Database	PostgreSQL + Spring Data JPA	Equipment, bookings, customers, orders
Auth	Spring Security + JWT	Protects the admin dashboard
Migrations	Flyway	Versioned database schema changes
Frontend	Vite + React + TypeScript	Public site and admin dashboard UI
Styling	Tailwind CSS	Fast, consistent visual design
Booking UI	Calendar/date-range component	Per-item availability display

Core Data Model

- **Equipment** — name, category, description, total quantity, price per unit per day, image
- **BookingRequest** — customer details, event date, overall status, created date
- **BookingItem** — links a booking to one piece of equipment: quantity, start/end date, unit price, line total, and its own status (pending / confirmed / cancelled)

Suggested Build Order

- **Phase 1** — Public site with equipment catalogue and quote/enquiry form
- **Phase 2** — Equipment/inventory model with admin CRUD
- **Phase 3** — Booking system with per-item dates and quantity-aware availability
- **Phase 4** — Admin dashboard: orders, booking status, payment confirmation